SwinFusion++: Face-Aligned Spatio-Temporal Swin Transformer for Deepfake Detection

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Abstract– Combining spatial and temporal representations does not establish that cross-attention improves deepfake detection. This study evaluates SwinFusion++, a face-aligned dual-stream detector integrating Swin-B and Video Swin-B through tubelet-aligned, local, bidirectional, gated hierarchical cross-attention (TLBG-HCA). Evaluation covers FaceForensics++ DeepFakes C23/C40, Celeb-DF v2, and an audited DeeperForensics-1.0 binary construction. Video-level comparisons distinguish documented training cohorts, individual models, homogeneous seed ensembles, and a single-seed frozen-backbone fusion experiment. Uncertainty analysis accounts for recorded dependency groups; frozen transfer retains source-trained weights, calibration, and thresholds. Fusion ensembles achieved ROC-AUCs of 0.9965, 0.9854, 0.9724, and 0.9990 across the four respective conditions, without consistent superiority over standalone encoders. Against the representative CNN–LSTM on C40, seed-1701 fusion improved AUC by 0.1846 (95% paired interval 0.1503–0.2165; Holm-adjusted $p = 0.0003999$). However, full TLBG-HCA and the capacity-matched no-exchange head had identical observed AUC and F1 in the controlled experiment; no incremental exchange advantage was established. Frozen transfer to the full DeeperForensics construction yielded AUC 0.8021 but fake recall only 13.41%; strict lineage filtering yielded AUC 0.7682 and recall 17.90%. Calibration limitations and additional computational cost further qualify practical interpretation. These findings support benchmark discrimination more strongly than exchange-specific benefit or reliable transfer, demonstrating why architectural performance, mechanism attribution, and operating-point validity require separate evidence.

Keywords– deepfake detection; Swin Transformer; Video Swin Transformer; cross-attention; spatio-temporal fusion; cross-dataset evaluation

1. Introduction

1.1. Context and Motivation

A deepfake detector must do more than recognize the artifacts of a familiar dataset: its decisions must remain informative when video quality and data source change. FaceForensics++ demonstrates the importance of evaluating facial manipulations under compression, while Celeb-DF challenges detectors with higher-quality face swaps that reduce conspicuous synthesis artifacts (Y. Li et al., 2020; Rössler et al., 2019). Strong discrimination within a benchmark is therefore only one part of the problem. Generalization to unfamiliar manipulations and resilience to ordinary post-processing are distinct requirements, as the LipForensics study illustrates (Haliassos et al., 2021). Benchmark success must therefore be distinguished from reliable detection under distribution shift.

This challenge motivates joint consideration of facial appearance and temporal behavior. Spatial analysis can examine localized image structure, whereas temporal analysis can reveal irregularities in how facial movements evolve (Haliassos et al., 2021; Rössler et al., 2019). Aggregating independent frame predictions cannot explicitly represent interactions between frames; conversely, using a video encoder does not establish that an additional spatial stream contributes useful information. A dual-stream detector must therefore justify both the complementarity of its representations and the mechanism used to combine them.

1.2. Problem Statement and Research Gap

Swin Transformer constructs multiscale image representations through hierarchical shifted-window attention (Liu et al., 2021); Video Swin Transformer extends this local-attention design to space and time (Liu et al., 2022). Their combination nevertheless presents a correspondence problem. In the configuration studied here, framewise Swin processes a 16-frame clip at 16 temporal positions, whereas Video Swin's stride-two tubelets produce eight. Intermediate features must be temporally aligned before corresponding local regions can exchange information. Combining only the final pooled representations provides a useful baseline, but leaves the value of interaction across earlier hierarchy levels untested.

Prior work establishes a basis for this inquiry. CAST couples CNN-derived spatial features with Transformer-based temporal representations through cross-attention (Thakre et al., 2026b). For the present dual-Swin design, the question is more specific: *does temporally aligned, multilevel exchange between Swin-B and Video Swin-B add discriminative value beyond the individual backbones, late concatenation, and a capacity-matched control without cross-attention?* This framing separates the usefulness of a dual-stream system from the incremental contribution of its exchange mechanism.

A second issue is the validity of the comparison. DeepfakeBench identifies inconsistent preprocessing, experimental settings, and evaluation protocols as sources of misleading performance comparisons (Yan et al., 2023). Accordingly, shared video-level evaluation is necessary but not sufficient: differences in training budgets, data partitions, and decision thresholds must remain visible. Published results obtained under incompatible protocols provide context, not a common leaderboard or a basis for paired statistical inference.

1.3. Objective

This study evaluates **SwinFusion++**, a face-aligned dual-stream detector combining Swin-B and Video Swin-B through Tubelet-Aligned, Local, Bidirectional, Gated Hierarchical Cross-Attention (TLBG-HCA). At four matched hierarchy

stages, learned temporal adapters align frame-indexed spatial features with video tubelets, after which local cross-attention exchanges information in both directions. Zero-initialized scalar residual gates regulate the added updates. Each stage contributes the two enhanced representations, their absolute difference, and their elementwise product to classification. Fusion is readout-only: these enhanced features do not replace the representations propagated through either backbone.

The general objective is to characterize the detector's discrimination, transfer behavior, calibration, and computational demands while distinguishing architectural performance from mechanism-specific evidence.

1.4. Scope and Contributions

Evaluation is restricted to visual, face-centered binary deepfake detection using FaceForensics++ DeepFakes C23/C40, Celeb-DF v2, and the study's audited DeeperForensics-1.0 binary construction. These benchmarks provide compression, higher-realism synthesis, and perturbation-oriented evaluation contexts, respectively (Jiang et al., 2020; Y. Li et al., 2020; Rössler et al., 2019). The project uses declared group-disjoint partitions, aligned facial clips, one prediction per analyzed video, and uncertainty estimates that account for shared source or identity groups. Frozen transfer carries the C23-trained model, calibration, and threshold to C40, Celeb-DF, and DeeperForensics without target-label fitting. Ranking metrics are interpreted alongside calibration, class-sensitive decision measures, and preprocessing coverage rather than accuracy alone.

The scope excludes audio, body-only synthesis, full-scene generation, and FaceForensics++ manipulation categories other than DeepFakes. Training-budget differences in some comparisons, class imbalance, concentrated test dependencies, and a single-seed frozen-backbone fusion ablation constrain attribution and generalization. In particular, the no-exchange control still combines both streams; it isolates cross-attention rather than all forms of fusion. Hyperparameter search and deployment validation are outside the completed evaluation. Within these boundaries, the study contributes an implementation-grounded dual-stream architecture, controlled comparisons of fusion alternatives, and an auditable assessment of performance and transfer. Its significance lies in establishing what the evidence supports about spatial–temporal integration, including its limitations, rather than equating benchmark success with forensic reliability.

2. Review of Related Literature

Deepfake detection literature offers competing explanations for generalization: recovering manipulation traces, learning facial dynamics, shaping training data, and combining representations. For SwinFusion++, both encoders observe the same RGB facial clips; different architectures do not guarantee independent evidence or complementary errors. The shortcut-learning account explains why successful predictors can fail when incidental correlations change (Geirhos et al., 2020). This review therefore distinguishes the performance of a complete detector from evidence that its cross-stream exchange contributes additional value.

2.1. Spatial Evidence, Frequency Cues, and Training Data

Xception separates spatial filtering from channel mixing, and its use in FaceForensics++ establishes an important frame-based reference (Chollet, 2017; Rössler et al., 2019). Averaging independent frame predictions does not explicitly model their order, but this limitation does not make spatial detectors intrinsically inadequate. Face X-ray emphasizes blending boundaries, while Self-Blended Images (SBI) constructs training examples by blending transformed versions derived from pristine images (L. Li et al., 2020; Shiohara & Yamasaki, 2022). Both challenge an architecture-only account of transfer: supervision and positive-example construction can encourage more generic inconsistency detection without adding a video encoder. Their evidence remains bounded by the manipulation traces available, particularly when blending boundaries are absent or obscured.

F3-Net investigates frequency-aware clues through decomposed image components and local frequency statistics (Qian et al., 2020). TSFF-Net combines spatial cues, including chrominance and edges, with wavelet-based frequency information (H. Zhang et al., 2024). Such spatial–frequency systems differ from spatial–temporal fusion; a shared two-stream label does not imply equivalent inputs or reasoning. Frequency sensitivity also does not establish invariance to compression or resizing. LAA-Net's artifact-oriented attention and feature pyramid further show that local evidence and multiscale integration are established concerns (Nguyen et al., 2024). The unresolved issue is not whether retaining detail is desirable, but whether a particular fusion arrangement improves discrimination beyond alternative supervision and simpler joint representations.

2.2. Temporal Modeling and Representation Learning

Convolutional–recurrent detection and attention-based convolutional LSTM processing establish temporal modeling before video Transformers (B. Chen et al., 2022; Güera & Delp, 2018). I3D supplies a related representational precedent through three-dimensional convolution for action recognition, rather than direct evidence of forensic transfer (Carreira & Zisserman, 2017). These approaches remain relevant comparators, but architecture, pretraining, and evaluation protocol must be distinguished when interpreting their reported performance.

Temporal-focused studies also question whether additional spatial capacity is necessarily helpful. FTCN restricts spatial convolutional kernels while retaining temporal filtering and a temporal Transformer (Zheng et al., 2021). This counterpoint motivates testing whether a spatial branch contributes useful evidence or reintroduces dataset-specific cues. LipForensics instead uses visual-speech pretraining and mouth dynamics; RealForensics develops representations through audio-visual self-supervision on real talking faces (Haliassos et al., 2021, 2022). Their transferable information is shaped by learning objectives and external supervision, not backbone topology alone.

AltFreezing alternates the freezing of spatial and temporal weights to encourage use of both cue families (Wang et al., 2023). It is distinct from fixing both encoders while com-

paring fusion heads: one changes representation learning, the other tests readouts of fixed representations. Together, these studies separate architectural access to temporal information from learned reliance on it and from its ability to transfer. Table 1 summarizes the competing explanations that inform the present controls.

2.3. Hierarchical Transformers and Cross-Stream Fusion

Scaled dot-product attention supplies a mechanism for token interaction (Vaswani et al., 2017). Swin Transformer uses patch merging and shifted local windows to construct hierarchical image features; Video Swin extends local attention to space and time (Liu et al., 2021, 2022). Within deepfake detection, ISTVT uses decomposed spatial–temporal attention and a self-subtract mechanism, while SFormer provides another spatiotemporal Transformer precedent (Kingra et al., 2024; Zhao et al., 2023). These studies preclude a novelty claim based simply on applying Transformers to facial video.

TALL arranges sampled frames into a thumbnail layout for an image backbone, including Swin (Xu et al., 2023). Temporal evidence can therefore be exposed through input design rather than a separate video stream. CrossViT exchanges information between image branches at different patch scales, while the action-recognition CAST uses bidirectional bottleneck cross-attention between spatial and temporal experts (C.-F. Chen et al., 2021; Lee et al., 2023). These are architectural precedents, not demonstrations of deepfake robustness.

The distinct deepfake CAST combines CNN spatial features with Transformer temporal features through cross-attention (Thakre et al., 2026b). Its revised preprint reports that a multiscale CNN-feature variant underperforms the main configuration (Thakre et al., 2026a, Section 5.5.4). This cautions against assuming that additional hierarchy levels improve fusion. The authors' representational-mismatch explanation motivates alignment, but does not establish that using two Swin encoders resolves mismatch automatically. A nonlinear late-concatenation classifier already integrates both descriptors; the stronger question is whether structured intermediate exchange contributes beyond that joint readout.

Hierarchy presents a related trade-off. Earlier stages preserve finer spatial detail, whereas later stages offer more aggregated representations. Combining stages may retain evidence lost through terminal pooling, but can also increase redundancy, trainable capacity, and sensitivity to dataset-specific detail. A multilevel design therefore requires controls that distinguish access to additional features from the particular mechanism used to exchange them.

2.4. Alignment, Residual Adaptation, and Readout

The present dual-Swin configuration retains 16 spatial frame positions but eight video tubelets, motivating a learned temporal adapter before corresponding local regions exchange information. Attention itself permits unequal query and key lengths; alignment here specifies the intended local correspondence, not a universal attention requirement (Liu et al., 2022; Vaswani et al., 2017). Matching temporal and spatial coordinates also does not guarantee identical semantics across encoders. Locality constrains interactions and attention-pair computation, but can exclude useful distant relationships and does not establish low total system cost.

ReZero's zero-initialized residual coefficient and LayerScale's small channelwise scaling provide related, distinct motivations for controlled residual adaptation (Bachlechner et al., 2021; Touvron et al., 2021). They do not establish that gating improves forensic transfer. In SwinFusion++, directly learned scalar gates are not input-dependent reliability probabilities. Enhanced features feed classification without replacing maps propagated between backbone stages. This readout-only design differs from recursively coupled expert exchange and does not imply gradient isolation when encoders are unfrozen. Likewise, absolute-difference and product features are candidate predictors, not uncertainty estimates. These distinctions delimit the hypothesis without attributing benefits to mechanisms that require separate interventions.

2.5. Generalization and Comparative Validity

FaceForensics++ supplies manipulation and compression conditions, Celeb-DF emphasizes higher-quality face swaps, and DeeperForensics-1.0 includes extensive perturbations (Jiang et al., 2020; Y. Li et al., 2020; Rössler et al., 2019). They are not a universal progression of difficulty. The present study retains only FF++ DeepFakes C23/C40 and an audited DeeperForensics binary construction, not every released manipulation or official benchmark protocol. Within-dataset testing, frozen source-to-target transfer, and target adaptation answer different questions. Deepfake-Eval-2024's revised preprint further documents degradation on collected contemporary media, but its acquisition process and multimodal scope do not establish prevalence or error rates for all online content (Chandra et al., 2026).

Preprocessing determines which evidence reaches a model. MTCNN addresses face detection/alignment, whereas FaceNet represents faces for recognition and clustering (Schroff et al., 2015; K. Zhang et al., 2016). Neither function validates authenticity. Cropping, track selection, and sampling can remove or alter useful evidence; failed inputs must remain visible through coverage. Image augmentation methods such as RandAugment also require explicit temporal-consistency choices when applied to video (Cubuk et al., 2020). Equal tensor dimensions alone do not establish equal observed video content.

DeepfakeBench identifies inconsistent preprocessing and experimental protocols as obstacles to reliable comparison (Yan et al., 2023). Shared test videos do not remove differences in pretraining, training budgets, or checkpoint selection. Structured-validation principles additionally require recorded source and identity dependencies to inform partitioning and uncertainty (Roberts et al., 2017). A capacity-matched no-exchange control reduces a competing explanation for fusion gains, but does not equalize all optimization dynamics. A frozen-backbone comparison narrows attribution while leaving end-to-end co-adaptation and training-seed sensitivity unresolved.

Evaluation must separate ranking, decisions, and probability quality. Precision-related measures depend on prevalence and the operating point (Saito & Rehmsmeier, 2015); temper-

Table 1. Literature synthesis for aligned dual-Swin fusion. The final column states this review's interpretation, not a common numerical ranking.

Evidence and examples	Contribution to present knowledge	Unresolved implication for this study
Artifact-oriented learning: Face X-ray; SBI	Blending supervision and synthetic training examples can support generalization.	Separate training-information benefits from architectural effects.
Temporal learning: FTCN; LipForensics; RealForensics	Temporal emphasis and learned facial dynamics offer alternatives to added spatial capacity.	Test complementarity rather than infer it from encoder names.
Alternative integration: TALL; CrossViT; both CAST models	Layout design and cross-attention provide established routes to combining representations.	Generic multiscale or bidirectional fusion is not itself novel.
SwinFusion++ (present study)	Tubelet-aligned local exchange with gated, hierarchical, readout-only fusion.	Compare against individual encoders, late concatenation, and capacity-matched no-exchange fusion.

ature scaling can adjust confidence without changing binary score ordering, and source calibration need not survive distribution shift (Guo et al., 2017; Ovadia et al., 2019). Nonsignificance also does not establish equivalence, which requires a justified margin and suitable analysis (Lakens, 2017). These principles support reporting uncertainty, coverage, and computational boundaries alongside accuracy rather than using one favorable metric as evidence of reliability.

2.6. Synthesis and Unresolved Gap

The literature supports conditional, not automatic, benefits from combining spatial and temporal representations. Artifact-oriented learning favors local detail; temporal-focused approaches caution against spatial shortcuts; synthetic and self-supervised training show that generalization gains need not originate in fusion. Multiscale and cross-attention precedents make the proposed mechanism plausible while leaving its incremental value uncertain.

The unresolved question is whether tubelet-aligned, local bidirectional exchange at matched Swin-B and Video Swin-B stages improves video-level discrimination or transfer beyond the individual encoders and strong joint-readout controls. SwinFusion++'s distinctive contribution is the specified combination of temporal alignment, hierarchical local exchange, scalar residual gates, and readout-only aggregation, not the invention of those components. The working hypothesis predicts useful cross-stream relationships beyond pooled or independent processing; rival explanations include redundant representations, extra capacity, pretraining, and unequal optimization budgets. Testing these alternatives permits positive, null, and condition-dependent findings to develop the thesis without equating benchmark performance with exchange-specific benefit or deployment readiness.

3. Theoretical Framework

This study synthesizes hierarchical representation learning, cross-attention, residual adaptation, and distribution-shift analysis into an account of *conditional representational complementarity*: aligned interaction may improve detection when it exposes label-relevant relationships beyond either encoder or a simpler joint readout. Complementarity, exchange benefit, and transferability are separate empirical claims.

3.1. Representational Complementarity

Swin Transformer supplies hierarchical image features, while Video Swin extends local attention to space and time (Liu et al., 2021, 2022). These architectures motivate retaining both localized appearance and temporally contextualized representations at matched scales. LipForensics provides evidence that learned facial dynamics can support generalization, whereas FTCN shows that restricting spatial processing can also improve transfer (Haliassos et al., 2021; Zheng et al., 2021). Together, these studies motivate a conditional account: extra spatial information can support relevant discrimination or introduce additional shortcuts.

Both streams observe the same RGB clips; their errors need not be independent. Different receptive fields and pretraining can expose complementary relationships or shared nuisance correlations. The shortcut-learning account explains why predictive relationships may fail outside their training distribution (Geirhos et al., 2020). Complementarity is therefore an empirical property of a trained system under specified conditions, not a theorem about the encoder names. A performance gain does not itself identify learned manipulation cues.

3.2. Aligned Exchange and Residual Adaptation

CrossViT and the action-recognition CAST motivate interaction before terminal pooling without establishing forensic effectiveness (C.-F. Chen et al., 2021; Lee et al., 2023). In the present design, a learned depthwise stride-two temporal adapter maps 16 spatial time indices to eight video-tubelet indices at each stage $i \in \{1, 2, 3, 4\}$. Let A_i and B_i denote corresponding spatial and video windows after alignment and channel projection. Shared coordinates supply a correspondence for exchange, not a guarantee of semantic equivalence.

For corresponding windows, the attention residuals are

$$\begin{aligned} A_i^a &= A_i + \gamma_i^S \, \mathcal{C}_i^{S \leftarrow V}(A_i, B_i), \\ B_i^a &= B_i + \gamma_i^V \, \mathcal{C}_i^{V \leftarrow S}(B_i, A_i). \end{aligned} \tag{1}$$

Here $\mathcal{C}$ denotes separately parameterized multi-head cross-attention with streamwise normalization, based on scaled dot-product attention (Vaswani et al., 2017). Both directions use the original normalized inputs in parallel. Each interaction set comprises 49 spatial tokens within a 7×7 window at one aligned tubelet index; temporal context comes from the video backbone and adapter, not unrestricted cross-attention across

all tubelets. Locality is therefore a constraint on possible interactions, whose usefulness remains testable.

ReZero motivates zero-initialized residual coefficients (Bachlechner et al., 2021). In SwinFusion++, the attention gates in Equation (1) and separate feed-forward gates are direct scalars: four per stage, initialized to zero. Initially the added paths vanish, while projection and classification remain active. This motivates cautious adaptation, not guaranteed convergence or useful exchange. Gate values are not input-dependent reliability probabilities.

The enhanced streams, their absolute difference, and their elementwise product enter normalized, mean/max-pooled stage readouts. These are candidate predictors, not uncertainty estimates. Fusion is readout-only: enhanced maps inform classification without replacing features passed to later backbone stages. This does not prevent gradients reaching unfrozen backbones.

3.3. Empirical Propositions and Competing Explanations

The *complementarity proposition* predicts useful joint discrimination beyond the standalone encoders. The stronger *exchange proposition* predicts an incremental advantage over a joint model that lacks cross-attention. For the controlled C23 frozen-backbone comparison, let F, L, and N denote full TLBG-HCA, late concatenation, and capacity-matched no-exchange fusion. Define

$$\begin{aligned} \Delta_{\text{exchange}} &= \text{AUC}(F) - \text{AUC}(N), \\ \Delta_{\text{pooled}} &= \text{AUC}(F) - \text{AUC}(L). \end{aligned} \tag{2}$$

These video-level contrasts use common test videos. Support for an exchange advantage requires a positive Δ_{exchange} with uncertainty addressed; $\Delta_{\text{pooled}} > 0$ alone is insufficient because stage access and readout capacity also differ. The N control retains alignment, hierarchy, gates, and difference/product readout, replacing cross-attention with independent self-attention. The contrast therefore concerns cross-stream attention versus that alternative, not fusion versus no fusion.

The rival explanation is that independent processing and joint readout already account for useful predictions. An unsupported exchange advantage limits the stronger proposition without disproving spatial–temporal integration. Equal metrics or nonsignificant differences do not establish equivalence (Lakens, 2017). The single-seed frozen-feature comparison also cannot establish training-run robustness or the benefit of end-to-end co-adaptation. These propositions organize the documented evaluation; they are not claims of prospective registration.

The *transfer proposition* asks whether useful relationships persist when the observation distribution changes. Source discrimination, target discrimination, and probability calibration are distinct outcomes. Temperature scaling can adjust confidence without improving binary score ordering, and source calibration need not survive shift (Guo et al., 2017; Ovadia et al., 2019). Frozen transfer therefore retains source-trained weights, calibration, and threshold. DeepfakeBench's standardization concerns further require training budgets, preprocessing, and evaluation units to remain visible when interpreting contrasts (Yan et al., 2023).

3.4. Research Questions

Five questions connect these propositions to the implemented comparisons:

RQ1: What within-dataset discrimination and seed-stability profile does SwinFusion++ exhibit relative to standalone Swin-B and Video Swin-B across the retained dataset conditions?

RQ2: How does SwinFusion++ compare with verified reproduced cross-family baselines, with protocol-qualified literature evidence kept separate from matched experimental comparisons?

RQ3: What differences in video-level ROC-AUC and F1 arise against the selected CNN–LSTM baseline under C23 and C40, and which comparisons support formal paired inference?

RQ4: Under fixed C23 backbone features and a common head-training budget, what incremental value does full TLBG-HCA provide over late concatenation and the capacity-matched no-exchange control?

RQ5: How do compression and frozen cross-dataset transfer affect discrimination, calibration, and thresholded decisions, and what seed-stability, preprocessing-coverage, and computational constraints qualify their interpretation?

3.5. Contributions and Boundaries of Inference

The study contributes an implementation-grounded configuration of aligned hierarchical dual-Swin fusion; a controlled distinction between system-level performance and exchange-specific evidence; and an auditable account of transfer, calibration, coverage, and cost. Its contribution is the specified combination and assessment, not the invention of attention or gating. The existing controls do not separately establish effects of alignment, locality, stage selection, or gate initialization; these require distinct interventions.

The scope remains the retained face-centered visual benchmarks. Unequal training cohorts, concentrated dependency groups, and single-seed ablation constrain attribution and generalization. Positive, null, and condition-dependent findings are informative only relative to their comparator, population, and training regime: the framework makes fusion's value an empirical question rather than an architectural assumption.

4. Methodology/Research Design

4.1. Design, Experimental Units, and Hypotheses

The study used a quantitative computational experimental design for face-centered binary deepfake detection. Architecture, fusion readout, and source–target condition were the experimental factors; the recorded training cohort constrained interpretation. A video was the prediction and evaluation unit, a 16-frame clip was the training input, and a dependency group represented related videos for splitting and resampling. Manipulated videos were positive ($y = 1$). A preprocessing failure remained in the coverage denominator without an invented prediction. Pre-existing benchmark videos and labels were used; no participants were recruited, surveyed, or recorded.

The five questions in Section 3 were operationalized through 36 principal model–condition–seed cells (three models, four conditions, and seeds 1701, 2203, and 3109); six reproduced-baseline cells; three controlled fusion-head cells; and 27 frozen-transfer evaluations using existing source checkpoints. RQ1 concerns principal discrimination and seed variation; RQ2 concerns verified cross-family baselines; RQ3 isolates the selected recurrent comparison; RQ4 concerns incremental exchange; and RQ5 concerns transfer, calibration, coverage, and cost. Blocked implementation cells were not counted as completed experiments.

The working hypothesis was that aligned hierarchical exchange could improve discrimination beyond individual encoders and simpler joint readouts. For an eligible comparison j and metric $m \in \{\mathrm{AUC}, F_1\}$, the observed contrast was $\widehat{\Delta}_{j,m} = m(A_j) - m(B_j)$, with statistical hypotheses $H_{0,j,m} : \Delta_{j,m} = 0$ and $H_{1,j,m} : \Delta_{j,m} \neq 0$. The implemented test was two-sided despite the directional scientific expectation. Complete-system comparisons do not isolate architecture from training, whereas RQ4 holds backbone features fixed. These hypotheses organize the documented analysis; no prospective registration or retrospective power calculation is claimed.

4.2. Materials, Partitioning, and Source Lineage

The retained materials were FaceForensics++ (FF++) DeepFakes C23/C40, Celeb-DF v2, and the project's DeeperForensics-1.0 binary construction (Jiang et al., 2020; Y. Li et al., 2020; Rössler et al., 2019). FF++ contributed 1,000 authentic and 1,000 manipulated logical videos per compression condition. Celeb-DF contained 890 authentic and 5,639 manipulated videos after excluding two exact notebook-checkpoint duplicates. DeeperForensics contained 11,000 manipulated videos across 11 retained variants and 1,000 authentic logical inputs reused from FF++ C23 originals. It is therefore neither the complete DeeperForensics release nor an intrinsically independent target domain. Audio, body-only edits, full-scene synthesis, and other FF++ manipulation categories were outside scope.

Read-only inventory manifests linked video identifiers, labels, relative paths, hashes, source/derivative relations, available identities, and acquisition metadata. FF++ followed its recorded source-pair split, shared across C23/C40. Celeb-DF used a custom identity-component-disjoint split targeting 70/15/15 proportions: assignments of four known components were enumerated, remaining provisional assignments retained, and deviations from target overall and class proportions minimized. DeeperForensics used recorded target/source-actor assignments. The realized counts in Table 2, not nominal proportions, define the experiment. Validation and test class imbalance was retained; only training sampling balanced classes.

Audits checked one row per logical video, valid partition labels, and disjoint recorded dependency, derivative, exact-hash, identity, and source groups within each condition. These checks support separation under available metadata, not discovery of every latent relationship (Roberts et al., 2017). The cross-dataset audit recorded 1,000 exact reuse edges between DeeperForensics authentic inputs and C23 originals. Of 2,412 DeeperForensics target-test rows, 2,052 had target lineage intersecting C23 development identifiers. Removing targets linked to C23 training/validation target or donor identifiers left a strict 360-video transfer subset, with 354 successfully analyzed. Full, overlap-qualified and strict views were reported separately. C23-to-Celeb-DF had no recorded exact development-to-target hash overlap, but lacked a verified cross-dataset identity correspondence; an identity-disjoint transfer claim is not made.

4.3. Face Processing, Sampling, and Augmentation

A frozen common RGB cache was constructed before training. For duration D, 64 candidate timestamps were placed uniformly from $0.02D$ to $0.98D$; PyAV decoded the first available frame at or after each request. Sampling was normalized by duration, not a fixed frame rate. The `facenet_pytorch` MTCNN detector used minimum face size 40, cascade thresholds $(0.6, 0.7, 0.7)$, scale factor 0.709, and accepted confidence at least 0.90 (K. Zhang et al., 2016). Detection batches were 16 frames, or 32 for frames of at least two million pixels.

The highest-confidence accepted face initialized a primary track. VGGFace2-pretrained InceptionResnetV1 embeddings supported association, not manipulation classification (Schroff et al., 2015). Candidates required cosine distance at most 0.55 or intersection-over-union (IoU) at least 0.10, and were ranked by cosine distance plus $0.15(1 - \mathrm{IoU})$. The ambiguity margin was 0.05 and embedding exponential-moving-average weight 0.90. Ambiguous or unmatched observations were skipped. Five-landmark similarity alignment used the fixed canonical 112-pixel template scaled by two to 224×224, linear interpolation, and reflected borders; the exact template and tracking implementation remain in the retained preprocessing configuration.

At least 48 accepted observations were required. Longer tracks were deterministically reduced to 48 evenly spaced observations and stored as unsigned 8-bit RGB banks. Evaluation used three disjoint 16-position anchors starting at 0, 16, and 32. Decode, face-confidence, track-association, insufficient-evidence, and cache-integrity failures remained explicit. No missing track was filled with synthetic evidence. Because failed detections and variable video duration alter temporal spacing, a 16-position input is not necessarily 16 consecutive frames.

A seeded hierarchical distributed sampler alternated labels, cycled through shuffled dependency groups, and drew a video within the chosen group. Draw counts were rounded for distributed sharding; sampling could repeat videos rather than visiting each exactly once. Hashes of seed, epoch, video identifier, and sampling purpose selected a training start among 0–32 and its augmentation. Horizontal flip probability was 0.5; shared clipwise intensity gain was uniform on $[0.92, 1.08]$ and additive bias on $[-6, 6]$, followed by clipping. Transforms were identical across a clip's frames, with no general RandAugment search. Principal tensors were $B \times 16 \times 3 \times 224 \times 224$, divided by 255 and normalized with means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Evaluation had no augmentation.

Table 2. Frozen partitions and analyzed test populations. Pairs are authentic/fake; train, validation, and test counts precede preprocessing.

Condition	Train	Validation	Test	Analyzed test	Groups
FF++ DeepFakes C23	720/720	140/140	140/140	140/140	70
FF++ DeepFakes C40	720/720	140/140	140/140	140/140	70
Celeb-DF v2	609/4,188	145/728	136/723	136/723	46
DeeperForensics binary	703/7,733	96/1,056	201/2,211	200/2,185	20

Groups refer to highest-level recorded test dependencies. Strict DeeperForensics transfer uses a separate 360-video manifest and 354 analyzed videos, not a fifth training condition.

4.4. Encoder and Fusion Specification

The spatial encoder was torchvision Swin-B with `IMAGENET1K_V1` weights; the video encoder was Video Swin-B with `KINETICS400_V1` weights (Liu et al., 2021, 2022). Registered checkpoint identities preceded matrix execution. The standalone spatial model averaged frame embeddings before a binary clip head; the video model classified its pooled video embedding. Both fusion branches observed the same normalized RGB clip, rather than separate sensing modalities.

At the four stages, channel widths were $(128, 256, 512, 1024)$ and spatial resolutions $(56, 28, 14, 7)^2$. Framewise features retained 16 positions; video tubelets retained eight. A learned depthwise convolution with kernel and stride $(2, 1, 1)$ aligned the spatial stream:

$$\widetilde{S}_{i,t,k} = a_{i,k,0} S_{i,2t,k} + a_{i,k,1} S_{i,2t+1,k} + b_{i,k}. \tag{3}$$

Separate channel projections used dimensions $d_i = (64, 128, 256, 256)$ and $(2, 4, 8, 8)$ attention heads, each of dimension 32. Nonoverlapping 7×7 windows contained 49 tokens at one aligned tubelet position. For queries Q, keys K, and values U, attention was softmax$(QK^\top/\sqrt{32})U$, normalized over keys (Vaswani et al., 2017). Spatial queries used video keys/values and vice versa. Separately parameterized directions consumed the original normalized windows in parallel. Fusion windows were not shifted and did not jointly attend across all tubelets.

The attention residuals follow Equation (1); each is followed by its own pre-normalized, gated $d_i \to 4d_i \to d_i$ feed-forward residual with GELU. Attention and feed-forward dropout were 0.10. Four directly learned scalar gates per stage were initialized to zero: no sigmoid or tanh was applied in the executed implementation. This specifies the actual residual mechanism rather than importing a bounded-gate interpretation from related work (Bachlechner et al., 2021).

After reversing the window partition, enhanced grids entered

$$\begin{aligned} E_i &= \mathrm{LN}_{4d_i}[\widehat{S}_i; \widehat{V}_i; |\widehat{S}_i - \widehat{V}_i|; \widehat{S}_i \odot \widehat{V}_i], \\ r_i &= W_i^r[\mathrm{mean}_{t,h,w}\, E_i; \mathrm{max}_{t,h,w}\, E_i] + b_i^r. \end{aligned} \tag{4}$$

The $8d_i$ pooled descriptor was projected to 256 dimensions per stage. Four readouts formed 1,024 features, followed by layer normalization, a 512-unit GELU layer, dropout 0.30, and a scalar logit. Fusion was readout-only: enhanced maps did not replace features passed between backbone stages, although gradients could reach unfrozen encoders. Difference/product channels were candidate predictors, not uncertainty estimates. The head had 6,506,897 parameters and the complete detector 180,889,105.

4.5. Comparison Models and Controlled Intervention

Xception supplied four seed-1701 reproduced baseline cells using the verified FF++ implementation and registered weights (Chollet, 2017; Rössler et al., 2019). Shared crops were bilinearly resized to 299×299 with `align_corners=False`, normalized by mean/standard deviation 0.5, and the 16 frame logits averaged. This common-cache adaptation differs from the original release's crop pipeline. Blocks 8–12 and the exit flow became trainable at zero-based epoch 4.

The two recurrent cells used ImageNet-initialized ResNet-18 embeddings (512 dimensions), a one-layer unidirectional LSTM (256 hidden units), and a final-timestep binary classifier with dropout 0.30. ResNet layer 4 was eligible for unfreezing at epoch 4; both retained checkpoints were selected at epoch 3. This was a representative CNN–LSTM, not an exact reproduction of every cited recurrent detector (B. Chen et al., 2022; Güera & Delp, 2018). I3D and SFormer remained blocked at the experiment freeze because compatible checkpoint conversion or pinned author-controlled implementation/weight provenance was unavailable. CAST and TSFF-Net were literature context, not numerical local reproductions.

RQ4 fixed the selected C23 seed-1701 standalone encoder checkpoints in evaluation/inference mode and trained three new heads on deterministically replayed stage features. Late concatenation normalized each pooled 1,024-dimensional descriptor, projected each to 512 with GELU, and classified their concatenation through a normalized 512-unit GELU/dropout-0.30 head (1,581,057 trainable parameters). The capacity-matched no-exchange head retained alignment, stages, gates, pooling, and difference/product readout, replacing MHA(A, B, B) by MHA(A, A, A) in each stream. Full TLBG-HCA and no-exchange each had 6,506,897 trainable parameters. Thus, RQ4 tests cross-attention versus independent attention with joint readout, not fusion versus no fusion. Feature addresses linked source hashes, split/cache hashes, video, epoch, start, and augmentation; fingerprints checked replay across alternatives.

4.6. Optimization, Checkpoint Selection, and Cohorts

All models minimized unweighted binary cross-entropy on clip logits; balancing occurred in the sampler rather than an additional class-weighted loss. AdamW used $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 0.05, and maximum global gradient norm 1.0 (Loshchilov & Hutter, 2019). New

modules used learning rate 10^{-4} and unfrozen pretrained modules 10^{-5}. Biases, normalization terms, scalar gates, and other one-dimensional parameters were exempt from decay. Warmup occupied 10% of the cohort's planned optimizer steps, followed by cosine decay. No auxiliary identity, flow, contrastive, or gate-regularization loss and no hyperparameter search were used.

Two GPU workers and accumulation gave effective batch size 16. The principal runtime changed from one clip/GPU with eight accumulation steps to four clips/GPU with two; validation used four. Bfloat16 autocasting was used. The amendment preserved batch size, not a claim of identical stochastic trajectories. Optimizer groups were rebuilt upon unfreezing while preserving state for previously active parameters. Earlier encoder stages stayed frozen. Table 3 retains the schedule distinctions essential to interpreting the results.

Checkpoint ranking maximized validation video ROC-AUC, then average precision, then lower binary cross-entropy, then earlier epoch. Patience reset required AUC improvement of at least 0.001, so checkpoint selection and stopping were not identical rules. The C23 correction froze unused terminal backbone normalization parameters outside the distributed loss graph and reset checkpoint selection at resumed epoch 4. Its amended stage-4/3 thresholds were already reached at that boundary; both later stages were then trainable. These eight-completed-epoch runs retained the resource-correction designation rather than being relabeled full-budget.

Baselines used the common loss/optimizer, a 15-epoch maximum, patience four, and at least six completed epochs; Xception used two clips/GPU with four accumulation steps and CNN–LSTM four with two. RQ4 used effective batch 16, head learning rate 10^{-4}, maximum eight epochs, patience three, and at least three completed epochs. All three heads completed four epochs and selected epoch 0. This single-seed, fixed-feature experiment does not test end-to-end co-adaptation. Principal ensembles used three same-cohort seeds except Celeb-DF fusion, whose primary ensemble used only full-budget seeds 1701/2203; resource-bounded seed 3109 was reported separately.

4.7. Video Decisions, Calibration, and Frozen Transfer

Three unaugmented anchor logits were averaged per video. Eligible seed ensembles then averaged video logits, not probabilities, votes, or seed metric values:

$$\bar{z}_v = \frac{1}{3K}\sum_{k=1}^{K}\sum_{c=1}^{3} z_{v,c,k}, \qquad p_v = \sigma(\bar{z}_v/T), \qquad \widehat{y}_v = \mathbb{I}[p_v \geq q^*]. \tag{5}$$

Here $K = 1$ for a single model. Temperature scaling minimized validation binary cross-entropy by bounded optimization of $\log T \in [-5, 5]$; sigmoid inputs were clipped to $[-80, 80]$ for stability (Guo et al., 2017). The validation threshold q^* maximized Youden's $J = \text{TPR} - \text{FPR}$, choosing the lowest finite threshold within the implementation's tie tolerance. Ensemble operating points were fitted on ensemble validation logits. ROC-AUC and average precision used logits; probability and decision metrics used the calibrated output. Calibration does not alter binary score ordering in exact arithmetic.

Blinded test manifests omitted label fields during inference; logits and hashes were sealed before joining labels for evaluation. Training guards excluded test-partition access. Frozen transfer retained C23 weights, ensemble membership, temperature, and threshold unchanged for C40, Celeb-DF, and DeeperForensics, with no target-label fitting. Target-trained models answered a separate within-dataset question. DeeperForensics full and strict lineage views were not pooled. Coverage was analyzed videos divided by manifest videos; inventory exclusions, preprocessing failures, lineage exclusions, and prediction-join failures remained distinct.

4.8. Outcomes, Uncertainty, and Statistical Eligibility

Video ROC-AUC was primary. The archived `pr_auc` field was average precision, not trapezoidal precision–recall area. Secondary outcomes were accuracy, positive-class precision/recall/F1, specificity, confusion counts, binary cross-entropy, Brier score, and calibration error. Undefined positive ratios used the implementation's zero-division value of zero. ECE used 15 equal-width fake-probability bins, weighted absolute differences between mean probability and observed fake frequency, ignored empty bins, and included probability one in the last bin. Brier score was mean $(p_v - y_v)^2$. Secondary descriptive checks added balanced accuracy and an always-fake reference: for fake prevalence π, its accuracy is π, F1 is $2\pi/(1+\pi)$, and balanced accuracy is 0.5. These contextualize imbalance without retroactively becoming primary endpoints (Saito & Rehmsmeier, 2015).

The uncertainty analysis sampled the highest-level dependency groups with replacement and retained all their videos with sampled multiplicities; metrics remained video-weighted. It used 5,000 valid bootstrap replicates, seed 481516, 14 workers with spawned NumPy substreams, and 2.5th/97.5th percentile limits. One-class draws were rejected and redrawn. Temperature and threshold were fixed, so intervals concern test-sample variation conditional on a fitted system, not retraining, checkpoint selection, or calibration refitting. Cluster concentration and conditioning on both classes limit interpretation (Roberts et al., 2017). The 70/46/20 group counts in Table 2 must not be confused with numbers of independent people.

Paired comparisons joined common video identifiers and checked labels/groups, then used identical bootstrap groups for both models. Separate marginal intervals were not subtracted. Permutation independently swapped complete prediction bundles (logit, probability, and class decision) within groups with probability 0.5, with no recalibration. For 5,000 swaps,

$$p = \frac{1 + \sum_{r=1}^{5000} \mathbb{I}\{|\Delta_r| \geq |\Delta_{\text{obs}}|\}}{5001}. \tag{6}$$

The plus-one procedure avoids zero Monte Carlo probabilities (Phipson & Smyth, 2010). It assumes exchangeability under the cluster-swap null, not merely equality of two marginal AUC values.

Of 58 candidate pairs, 31 were protocol-eligible and 27 exploratory. Holm correction covered three fixed families: 56 metric-level internal hypotheses, two C40 seed-1701 fusion–

Table 3. Training cohorts retained in all comparisons. Epoch indices are zero-based; maximum and completed epochs are not interchangeable.

Cohort	Cells and schedule	Interpretation
Full-budget principal	23 cells: C23 standalone models; all C40 models; Celeb-DF standalone models and fusion seeds 1701/2203. Maximum 30 epochs; unfreeze stages 4/3 at epochs 4/8; patience 6; minimum 8 completed epochs.	Eligible within-cohort comparisons; realized durations can differ.
Fresh resource-bounded	10 cells: all nine DeeperForensics principal runs and Celeb-DF fusion seed 3109. Four epochs; unfreeze stages 4/3 at epochs 1/2.	Matched DeeperForensics cohort; the Celeb-DF run remains separate.
C23 fusion correction	Three runs: epochs 0–3 followed by corrected epochs 4–7, with checkpoint selection reset at the boundary.	Eight completed epochs; cross-cohort architecture/transfer differences are exploratory.
Reproduced baselines	Six seed-1701 cells; maximum 15 epochs, patience 4, minimum 6 completed; architecture-specific unfreezing.	Verification does not match their capacity or budget to principal models.
Frozen-feature heads	Three C23 seed-1701 heads; maximum 8 epochs, patience 3, minimum 3 completed. All completed 4 and selected epoch 0.	One controlled head-training regime, not a multi-seed end-to-end experiment.

All cohorts used effective batch size 16. AUC patience improvement was 0.001. The resource-correction label follows the recorded amendment rather than the run-name suffix.

CNN–LSTM hypotheses, and four RQ4 full-versus-control hypotheses (Holm, 1979). Effect intervals were not multiplicity-adjusted; significance used the adjusted probabilities, with 0.05 as the reporting threshold. C23 fusion–CNN–LSTM and fusion-versus-standalone transfer contrasts remained exploratory because source cohorts differed. The eligible C40 recurrent comparison still compares systems with different nominal budgets. Equal metrics or nonsignificance did not establish equivalence; no equivalence margin or test was supplied (Lakens, 2017).

4.9. Computational Measurement and Audit Trail

The recorded platform used two RTX 5090 GPUs, Python 3.12, PyTorch 2.12.1/CUDA 13.0, and torchvision 0.27.1 in the locked training environment. Inference profiling used evaluation/inference mode, bfloat16, one clip per worker, 20 warmups, 100 timed iterations, and CUDA synchronization. The first eligible input was reused; video timing included three sequential anchors. Latency remained a per-device measure, while dual-GPU throughput used combined completions divided by the slower worker's elapsed time. Decoding, face processing, host-to-device preparation, and seed-ensemble execution were excluded. Peak allocated/reserved memory, parameter counts, and checkpoint size were separate measurements.

Immutable identifiers connected source inventory, split, cache policy, configuration, pretrained weights, training amendments, selected checkpoint, and predictions. The review archive retained code, configurations, manifests, reports, and predictions, but not raw videos, complete caches, or large learned checkpoints needed for independent retraining. Numerical verification during manuscript preparation recalculated point metrics from saved predictions and checked Holm arithmetic; stored bootstrap intervals and permutation runs were not regenerated. This is an auditable evidence record, not a claim that training or benchmark labels were independently revalidated.

5. Presentation and Discussion of Results

5.1. Evidence Base and Analyzed Population

The results support a narrower conclusion than universal superiority: SwinFusion++ achieved strong within-dataset discrimination and a positive comparison against the selected C40 recurrent baseline, but did not consistently exceed its constituent Transformers, demonstrate an incremental cross-attention benefit in the controlled experiment, or preserve its source operating behavior under transfer. The distinction between these findings is central to assessing the thesis. A useful complete detector and a demonstrably useful exchange mechanism are different empirical achievements.

All 280 C23, 280 C40, and 859 Celeb-DF test videos produced valid predictions. The DeeperForensics construction yielded 2,385/2,412 analyzed test videos (98.88%); its strict transfer subset yielded 354/360 (98.33%). Analyzed class counts were 140 authentic/140 fake for each FF++ condition, 136/723 for Celeb-DF, 200/2,185 for full DeeperForensics, and 30/324 for the strict subset. These denominators, rather than all-partition inventory totals, apply to the reported performance. They also explain why accuracy and fake-class F1 cannot be interpreted without prevalence.

For context, all-partition preprocessing succeeded for 1,989/2,000 C23, 1,988/2,000 C40, 6,526/6,529 Celeb-DF, and 11,847/12,000 DeeperForensics videos. The last condition lost 153 videos overall (150 fake and three authentic), but only 27 from the held-out test set. Failures outside the test set must not be subtracted again from its denominator. Likewise, lineage filtering is a change of target population, not a detection failure. The high coverage establishes the reach of the retained face-processing policy on these inputs; it does not validate performance on videos without a usable primary face.

During preparation, saved predictions reproduced the stored ROC-AUC, average precision, accuracy, F1, Brier score, coverage, and counts for 72 individual evaluations and 21 ensembles. Three strict-transfer ensemble views were checked separately by applying their retained manifest identifiers, and Holm

adjustments were checked against the saved raw probabilities. This establishes numerical consistency, not independent retraining or label validation. Confidence intervals and permutation probabilities below are the retained dependency-aware analysis. New class-balance and cluster-composition checks are descriptive and do not replace that analysis.

5.2. Within-Dataset Discrimination and Operating Decisions

Table 4 presents homogeneous mean-logit ensembles. AUC summarizes ranking, whereas F1 and accuracy use each model's validation-derived operating point. On C23, SwinFusion++ obtained AUC 0.9965, F1 0.9712, and accuracy 97.14%. Video Swin-B had the largest observed AUC (0.9972), while Swin-B had higher F1 (0.9856) and accuracy (98.57%). Thus, the fusion model's improvement in AUC over Swin-B did not translate into improvement in fixed-threshold decisions. Because C23 fusion used the correction cohort while the standalone encoders used the full-budget cohort, these are descriptive system comparisons, not controlled evidence that fusion changed performance by the observed amount.

C40 provides a same-principal-cohort comparison. Fusion reached AUC 0.9854, below Swin-B (0.9917) and Video Swin-B (0.9930), and its F1 of 0.9420 was also below both standalone values. All three remained strong rankers on the retained DeepFakes population. The result does not support a general claim that introducing an additional spatial–temporal stream improves compressed-video detection. It instead shows that the principal dual-stream system can be less effective than a constituent encoder under the evaluated training procedure. Differences between C23 and C40 alone cannot be attributed solely to compression sensitivity because those conditions also used different fusion training cohorts.

On Celeb-DF, fusion's two-member full-budget ensemble reached AUC 0.9724, compared with 0.9824 for Swin-B and 0.9749 for Video Swin-B. Nevertheless, fusion obtained higher fake-class F1 (0.9561) and accuracy (92.78%) than either standalone ensemble. This is not a contradiction: ranking and the chosen decision boundary measure different properties. Fusion identified 675 of 723 fake videos, with 14 false positives among 136 authentic videos; Swin-B identified 649 fakes with only four false positives. Fusion's greater fake recall therefore accompanied weaker authentic specificity. Its balanced accuracy was 91.53%, versus 93.41% for Swin-B. The favorable accuracy/F1 result on a fake-majority test set should consequently not be described as dominance across operating requirements.

In the DeeperForensics construction, all three principal ensembles were near ceiling in AUC: 0.9989 for Swin-B, 0.9989 for Video Swin-B, and 0.9990 for fusion. The fusion–Swin AUC difference was approximately 0.0001, not a practically established advantage. At the retained thresholds, Swin-B had F1 0.9940 and accuracy 98.91%, compared with fusion's 0.9844 and 97.19%. Fusion made no false-positive decisions on the 200 analyzed authentic videos but missed 67 of 2,185 fake videos; Swin-B missed 25 and produced one false positive. These counts make the trade-off visible without turning zero observed false positives into a guarantee of zero population risk.

Across the four conditions, no model had the largest observed value for every metric. The strongest evidence for a useful decision function is therefore the joint profile of ranking, errors, and uncertainty, not selection of a favorable metric after seeing the outcome. Near-ceiling AUC also limits how informative small numerical separations can be, particularly when the test population contains few independent groups. RQ1 is answered by condition-dependent performance, not by an across-the-board fusion advantage.

5.3. Seed Variation and Formal Internal Comparisons

Table 5 separates variation across training runs from uncertainty over test observations. For fusion, same-cohort seed AUC was 0.9959 ± 0.0006 on C23, 0.9822 ± 0.0014 on C40, 0.9684 ± 0.0006 for the two full-budget Celeb-DF seeds, and 0.9990 ± 0.0001 on DeeperForensics. These standard deviations describe only two or three runs on the same split; they do not quantify variation over new datasets, identities, or alternative training recipes. The low Celeb-DF AUC of 0.7740 for resource-bounded seed 3109 belongs to another cohort and was not averaged into the full-budget result.

The ensembles are separate fitted decision systems, not arithmetic averages of these seed metrics. For example, full-budget Celeb-DF fusion had mean seed F1 0.9326 and ensemble F1 0.9561, whereas DeeperForensics fusion had mean seed F1 0.9904 and ensemble F1 0.9844. Averaging logits and fitting the ensemble operating point can change which videos cross the threshold. The second example is direct evidence that ensembling did not uniformly improve thresholded performance. The archive does not separate the contribution of score averaging from operating-point fitting, so neither component alone is assigned responsibility.

The internal Holm family contained 56 metric-level tests. At an adjusted probability below 0.05, two rejected null hypotheses were both unfavorable to the video-containing systems: on Celeb-DF seed 2203, fusion minus Swin-B AUC was -0.0152 (95% paired interval $[-0.0198, -0.0119]$), and Video Swin-B minus Swin-B was -0.0098 ($[-0.0131, -0.0072]$); each had $p_{\text{Holm}} = 0.0112$. These are specific seed-level comparisons, not significance labels for Table 4's ensembles. No eligible internal test established a positive fusion AUC/F1 advantage at that adjusted threshold. Non-rejection in the remaining tests is not evidence that their models are equivalent.

Table 6 places those negative results alongside the principal positive baseline result and the controlled fusion contrasts. This avoids treating statistical analysis as a search only for favorable findings. The intervals are percentile intervals and the probabilities are family-adjusted swap-test outputs; an interval excluding zero need not imply rejection after multiplicity correction. Their interpretation is further limited by the cluster structure discussed below.

5.4. Reproduced Cross-Family Baselines

The verified local baselines are reported in Table 7. Xception reached AUC 0.7725 on C23 and 0.8109 on C40; the representative CNN–LSTM reached 0.7876 and 0.7984. The princi-

Table 4. Within-dataset principal ensembles: video-level discrimination, retained operating decisions, and training cohort.

Condition	Model	K	Cohort	AUC	95% AUC interval	Fake F1	Acc. (%)
C23	Swin-B	3	F	0.9930	[0.9824, 1.0000]	0.9856	98.57
C23	Video Swin-B	3	F	0.9972	[0.9923, 0.9999]	0.9784	97.86
C23	SwinFusion++	3	C	0.9965	[0.9921, 0.9994]	0.9712	97.14
C40	Swin-B	3	F	0.9917	[0.9817, 0.9987]	0.9712	97.14
C40	Video Swin-B	3	F	0.9930	[0.9864, 0.9979]	0.9577	95.71
C40	SwinFusion++	3	F	0.9854	[0.9727, 0.9952]	0.9420	94.29
Celeb-DF	Swin-B	3	F	0.9824	[0.9785, 0.9835]	0.9433	90.92
Celeb-DF	Video Swin-B	3	F	0.9749	[0.9710, 0.9770]	0.9414	90.57
Celeb-DF	SwinFusion++	2	F	0.9724	[0.9680, 0.9746]	0.9561	92.78
DeeperForensics	Swin-B	3	R	0.9989	[0.9974, 0.9999]	0.9940	98.91
DeeperForensics	Video Swin-B	3	R	0.9989	[0.9970, 0.9999]	0.9861	97.48
DeeperForensics	SwinFusion++	3	R	0.9990	[0.9970, 1.0000]	0.9844	97.19

K is the number of same-cohort seed members; F = full-budget, R = fresh four-epoch resource cohort, C = C23 corrected continuation. Intervals are stored cluster-percentile intervals. No ensemble rank is a paired significance claim. The analyzed denominators are those in Table 2.

Table 5. Fusion seed means and sample standard deviations within each retained cohort.

Condition	K	AUC	Fake F1
C23	3	0.9959 ± 0.0006	0.9703 ± 0.0071
C40	3	0.9822 ± 0.0014	0.9347 ± 0.0032
Celeb-DF	2	0.9684 ± 0.0006	0.9326 ± 0.0194
DeeperForensics	3	0.9990 ± 0.0001	0.9904 ± 0.0052

Celeb-DF includes seeds 1701/2203 only. Resource seed 3109 has AUC 0.7740 and F1 0.8191 and is not part of that mean. SD is training-run variation, not a test-set confidence interval.

pal Transformer systems had substantially larger point AUCs on those test sets. This supports a comparison against these particular reproduced systems, not a statement that convolutional or recurrent methods are intrinsically unsuitable. The baselines differ from the principal models in pretrained information, nominal epoch budget, unfreezing schedule, and capacity, despite sharing accepted facial inputs and video-level evaluation.

The protocol-eligible RQ3 test compared seed-1701 C40 fusion, not its three-member ensemble, with the selected CNN–LSTM on 280 common videos in 70 dependency groups. Fusion AUC was 0.9830 versus 0.7984, giving $+0.1846$ (95% interval $[0.1503, 0.2165]$). F1 was 0.9382 versus 0.7515, giving $+0.1867$ ($[0.1503, 0.2201]$). Both adjusted probabilities were 0.0003999. This is the clearest retained positive paired comparison. It remains a difference between the specified trained systems: eligibility under the archived analysis does not erase their distinct budgets or establish a causal architecture-only effect. The C23 recurrent contrast remained exploratory and is not assigned an equivalent formal claim.

Two baseline outcomes demonstrate why F1 alone is insufficient. On Celeb-DF, Xception's F1 of 0.9140 accompanied predictions of fake for all 859 videos: it recovered every fake but rejected every authentic input, giving balanced accuracy 0.5000. Nonconstant scores still produced AUC 0.6480. On DeeperForensics, Xception achieved AUC 0.9978 yet F1 only 0.5821 and accuracy 46.00%, with 1,288 false negatives and no false positives. Strong ordering can therefore coexist with an unsuitable retained operating point, even without changing dataset between training and testing. The recorded temperature/threshold procedure is part of the evaluated system, not an optional correction applied afterward using test labels.

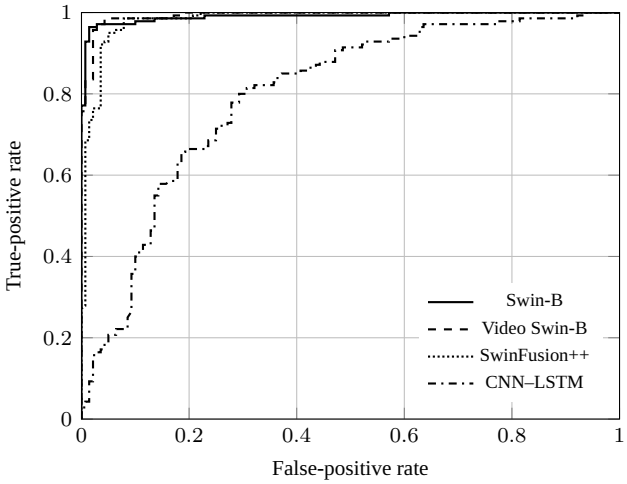


Figure 1. C40 seed-1701 ROC curves reconstructed from the stored video logits on 280 common test videos. The recurrent comparison concerns these single models, not the principal ensembles. Curves summarize all thresholds, whereas reported decisions retain validation-selected thresholds.

I3D and SFormer contributed no verified local numerical results; their blocked cells are not counted as detection failures. RQ2 is consequently supported by four Xception and two representative recurrent evaluations, with other methods supplying literature context only. Figure 1 visualizes the seed-1701 C40 scores used for that setting, rather than borrowing published curves from a different evaluation population.

5.5. What the Controlled Fusion Experiment Establishes

The C23 frozen-backbone comparison is the direct test of the exchange proposition in Section 3. Table 8 shows that full TLBG-HCA and the capacity-matched no-exchange head both obtained AUC 0.9964, F1 0.9857, and accuracy 98.57%. Late concatenation obtained AUC 0.9952, F1 0.9892, and accuracy 98.93%. Full minus late AUC was $+0.0012$ (95% paired inter-

Table 6. Selected protocol-eligible paired effects, including every metric-level rejection at adjusted $p < 0.05$ and all RQ4 contrasts.

Comparison (A − B)	Metric	Effect	95% paired interval	p_{Holm}
Celeb-DF: fusion − Swin-B (seed 2203)	AUC	-0.0152	[-0.0198, -0.0119]	0.0112
Celeb-DF: Video Swin-B − Swin-B (2203)	AUC	-0.0098	[-0.0131, -0.0072]	0.0112
C40: fusion − CNN–LSTM (seed 1701)	AUC	+0.1846	[+0.1503, +0.2165]	0.0003999
C40: fusion − CNN–LSTM (seed 1701)	F1	+0.1867	[+0.1503, +0.2201]	0.0003999
C23 fixed features: full − late concatenation	AUC	+0.0012	[-0.0005, +0.0040]	1.0000
C23 fixed features: full − late concatenation	F1	-0.0035	[-0.0141, +0.0075]	1.0000
C23 fixed features: full − no-exchange	AUC	+0.0000	[+0.0000, +0.0000]	1.0000
C23 fixed features: full − no-exchange	F1	+0.0000	[+0.0000, +0.0000]	1.0000

Internal, recurrent, and RQ4 families contain 56, 2, and 4 hypotheses, respectively. Intervals are not multiplicity-adjusted; probabilities are. Internal Celeb-DF effects are qualified by concentrated cluster support. RQ4 uses seed 1701 and fixed features; recurrent effects use individual models, not an ensemble. The other 54 internal tests did not reject at adjusted $p < 0.05$.

Table 7. Verified local baseline results (seed 1701).

Condition	Model	AUC	Fake F1	Acc. (%)
C23	Xception	0.7725	0.6886	69.64
C40	Xception	0.8109	0.7429	74.29
Celeb-DF	Xception	0.6480	0.9140	84.17
DeeperForensics	Xception	0.9978	0.5821	46.00
C23	CNN–LSTM	0.7876	0.7452	71.43
C40	CNN–LSTM	0.7984	0.7515	70.71

Xception uses the shared aligned cache; CNN–LSTM is the representative ResNet-18/LSTM implementation. Distinct nominal budgets prevent an architecture-only interpretation. No I3D or SFormer local score is asserted.

Table 8. Controlled fusion-head results on 280 C23 videos.

Readout	Head M	AUC	Fake F1	Acc. (%)
Late concatenation	1.58	0.9952	0.9892	98.93
No-exchange	6.51	0.9964	0.9857	98.57
Full TLBG-HCA	6.51	0.9964	0.9857	98.57

All heads share fixed seed-1701 encoders and a four-completed-epoch schedule. Head M denotes millions of trainable head parameters. Full/no-exchange retain different logits despite identical AUC, F1, and class decisions.

val $[-0.0005, 0.0040]$), while the F1 difference was -0.0035 ($[-0.0141, 0.0075]$). All four RQ4 adjusted probabilities were 1.0000. No incremental AUC or F1 advantage over either control was established in this experiment.

The full-versus-no-exchange finding is particularly informative because both heads had 6,506,897 trainable parameters, the same selected fixed encoders, and the same addressed training evidence. It weakens an exchange-specific explanation of the observed discrimination in that regime. However, the no-exchange model still jointly classified both streams, their absolute difference, and their product; independent self-attention replaced cross-attention. The result does not test whether all fusion is unnecessary, whether temporal information is unused, or whether either encoder alone contains all useful information.

Directly aligned predictions had identical class decisions on all 280 test videos, but the maximum absolute logit difference was 0.0208333 and the maximum calibrated-probability difference approximately 0.0005716. Checkpoint hashes differed. Equal reported metrics therefore did not arise from a demonstrated identity of models or outputs. The zero-width stored AUC/F1 effect intervals describe the empirical contrast generated by this sample and resampling scheme; they are not an equivalence test, nor evidence that a practically meaningful difference is impossible in another sample or training run (Lakens, 2017).

Several explanations remain compatible with the evidence. Frozen representations may already make the C23 distinction nearly separable, leaving little room for the exchange head. The streams may provide redundant information, or the common short head-training schedule may not exploit the added interaction. All three heads selected their epoch-0 checkpoint and completed four epochs, which makes later training an unconvincing basis for attributing their retained performance to progressively learned exchange. These are candidate explanations, not identified mechanisms: gate trajectories, attention interventions, and controlled retraining were not available to distinguish them. Nonzero or zero gates cannot be inferred from equal AUC alone.

Late concatenation used about one quarter of the trainable head parameters (1.58 million versus 6.51 million) while retaining comparable observed ranking and slightly higher F1. This identifies a substantive simpler control, not a definitive deployment recommendation: its total system still includes both encoders, and its AUC/F1 differences were not significant under the retained procedure. A future test of end-to-end co-adaptation, additional seeds, alignment variants, or stage ablation would answer questions that this single-seed readout experiment does not. RQ4 therefore does not support an incremental AUC/F1 improvement within this narrow, reproducible intervention.

5.6. Frozen Transfer: Ranking Does Not Ensure Useful Decisions

Table 9 keeps the C23-trained weights, source operating points, and ensemble membership fixed. Source-only transfer is stricter than evaluating target-trained systems, and source-cohort differences make fusion-versus-standalone comparisons descriptive. On C40, fusion transferred with AUC 0.9686 ($[0.9522, 0.9829]$), but F1 fell to 0.7686 and accuracy to 81.07%. It produced only one false positive among 140 authentic videos while missing 52 of 140 fake videos. High precision (0.9888) thus coexisted with recall 0.6286. The principal C40-trained fusion ensemble's F1 of 0.9420 does not demonstrate that the source-trained system can retain that operating behavior without target information.

Table 9. Frozen C23-source ensembles: unchanged source weights, temperature, threshold, and three-member composition.

Target/view	Source model	AUC	95% AUC interval	Fake F1	Acc. (%)	Recall (%)
C40	Swin-B	0.9719	[0.9532, 0.9878]	0.8664	88.21	76.43
C40	Video Swin-B	0.9440	[0.9190, 0.9677]	0.7182	77.86	56.43
C40	SwinFusion++	0.9686	[0.9522, 0.9829]	0.7686	81.07	62.86
Celeb-DF	Swin-B	0.9052	[0.8958, 0.9118]	0.9315	88.24	95.02
Celeb-DF	Video Swin-B	0.9045	[0.8901, 0.9138]	0.9289	88.24	91.29
Celeb-DF	SwinFusion++	0.8730	[0.8549, 0.8864]	0.8821	81.49	82.30
DeeperForensics (full)	Swin-B	0.8002	[0.7663, 0.8350]	0.3555	28.09	21.65
DeeperForensics (full)	Video Swin-B	0.7508	[0.7175, 0.7898]	0.2110	19.12	11.81
DeeperForensics (full)	SwinFusion++	0.8021	[0.7764, 0.8317]	0.2362	20.55	13.41
DeeperForensics (strict)	Swin-B	0.7767	[0.6978, 0.8593]	0.3636	28.81	22.22
DeeperForensics (strict)	Video Swin-B	0.7465	[0.6885, 0.8150]	0.3029	24.58	17.90
DeeperForensics (strict)	SwinFusion++	0.7682	[0.7148, 0.8282]	0.3029	24.58	17.90

Analyzed/manifest denominators: C40 280/280; Celeb-DF 859/859; DeeperForensics full 2,385/2,412; strict 354/360. Fusion source checkpoints belong to the correction cohort; standalone sources are full-budget. Cross-cohort ranks remain descriptive. Full and strict target views represent different populations; neither target operating point was refitted.

On Celeb-DF, transferred fusion AUC was 0.8730 ($[0.8549, 0.8864]$), below the standalone source ensembles' 0.9052 and 0.9045. Its accuracy of 81.49% and F1 of 0.8821 were below the always-fake references of 84.17% and 0.9140 on this particular 723/859-fake population. Those reference comparisons invalidate an unqualified high-accuracy success claim, but do not imply absence of useful discrimination. Fusion correctly accepted 105 authentic videos and detected 595 fakes, with 31 false positives and 128 false negatives; balanced accuracy was 79.75%, rather than the reference's 50%. Specificity was 77.21% and fake recall 82.30%.

The competing operating points illustrate the distinction. Transferred Swin-B had higher Celeb-DF accuracy (88.24%) and recall (95.02%) but lower specificity (52.21%) and balanced accuracy (73.61%). Video Swin-B also had 88.24% accuracy, with recall 91.29%, specificity 72.06%, and balanced accuracy 81.67%. These differences describe which errors each fixed system made on the same target population. They do not identify the optimal trade-off for another class prevalence or cost of error. Choosing a new threshold after inspecting these target labels would change the experiment into target-informed adaptation and is not done here.

Full-view DeeperForensics transfer exposed the most severe operational failure. Fusion maintained AUC 0.8021 ($[0.7764, 0.8317]$), yet obtained F1 0.2362 and accuracy 20.55%. It missed 1,892 of 2,185 fake videos, with recall only 13.41%, while rejecting three of 200 authentic videos. Balanced accuracy was 55.95%. Precision of 0.9899 is not reassuring in isolation: the system flagged very few positives in a test set where 91.61% of videos were fake. The always-fake accuracy/F1 references were 91.61%/0.9562. Conversely, an always-authentic rule would obtain only 8.39% accuracy. Neither trivial rule supplies useful discrimination; together they reveal how strongly the reported accuracy depends on the direction of errors under this prevalence.

Strict lineage filtering did not restore reliable transfer. On the 354 analyzed strict videos, fusion AUC was 0.7682 ($[0.7148, 0.8282]$), F1 0.3029, and accuracy 24.58%. Its confusion counts were 29 true negatives, one false positive, 266 false negatives, and 58 true positives. There were 17 recorded dependency groups. This smaller population removes the audited development-lineage intersection but is not a random subsample of the full target. The full-to-strict AUC change must therefore not be described as a causal estimate of leakage inflation: prevalence, identities, and other composition may also differ. Both views are needed to expose the lineage qualification and the persistence of weak decisions.

Figure 2 places fusion's ranking and threshold measures on the same target axes without treating them as interchangeable. AUC remains informative even where recall deteriorates sharply. A mismatch between source-trained score distributions and the retained threshold is consistent with this pattern; the experiment does not identify whether the underlying cause is synthesis variation, acquisition, preprocessing, source-specific cues, or their combination. In particular, the fall from near-ceiling target-trained DeeperForensics AUC to materially lower source-only AUC shows that the difficulty is not solely a threshold choice. Retuning the threshold could not, by itself, recover the lost score ordering.

The defensible RQ5 conclusion is that transfer is condition-dependent and weakest for the lineage-qualified DeeperForensics construction. Compression transfer, cross-dataset transfer, and strict source-lineage transfer answer different questions. None of these observations establishes behavior on every present-day deepfake generator or on audio and full-scene synthesis outside the retained visual task.

5.7. Calibration and Uncertainty Limits

Table 10 reports seed-1701 calibration to avoid attaching individual operating points to ensemble metrics. In each of the four within-dataset fusion evaluations, temperature scaling slightly reduced Brier score but increased ECE. On Celeb-DF, for example, Brier fell from 0.1218 to 0.1203 while ECE rose from 0.1279 to 0.2447, with validation-fitted temperature 5.8923. This is not evidence that one calculation must be wrong: Brier evaluates squared probabilistic error, whereas ECE aggregates

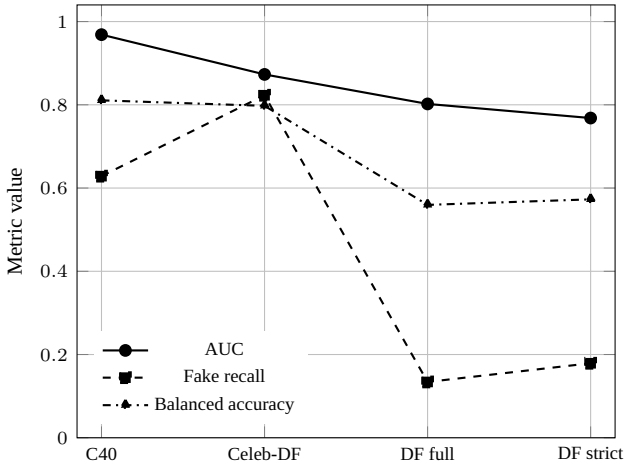


Figure 2. Frozen-transfer fusion ensemble profile. Ranking, fake recall, and balanced accuracy diverge under a fixed source operating point. Lines connect target categories for readability, not a temporal trajectory. DF denotes the DeeperForensics construction; full and strict views have different populations. Balanced accuracy is secondary descriptive analysis; AUC intervals appear in Table 9.

absolute discrepancies over probability bins. They also differ from the validation binary-cross-entropy objective used to choose temperature.

The source-only seed-1701 temperature 1.8251 and threshold 0.6629 were carried unchanged to all three targets. On DeeperForensics, scaling reduced Brier from 0.6362 to 0.5815 and ECE from 0.6815 to 0.6632, but the remaining errors were large. Relative improvement after a transformation is not sufficient evidence of adequate probability quality. This behavior is consistent with the wider finding that source calibration does not guarantee calibration after distribution shift (Ovadia et al., 2019). It also illustrates why temperature scaling, despite its established utility (Guo et al., 2017), should not be presented as a solution to the transfer problem demonstrated here.

The dependence structure is a more fundamental qualification than the number of printed decimal places. FF++ had 70 groups across 280 test videos; DeeperForensics had 20 across 2,385. Celeb-DF had 46 groups, but the largest contained 814/859 videos (94.76%). A descriptive inspection of the saved group labels further showed that this component contained all 723 fake videos and 91 authentic videos; the remaining 45 videos were authentic. Thus, a cluster bootstrap draw excluding the dominant component contains only one class and is rejected by the implemented procedure. The resulting valid draws necessarily retain that source of fake-class support.

Consequently, a narrow Celeb-DF interval cannot be read as evidence of broad, independently replicated fake-identity generalization. Resampling the dominant component with varying multiplicity does not create new manipulated components, and the inferential design conditions on this unusually concentrated support. The same limitation qualifies the statistically detectable negative seed-2203 comparisons rather than being invoked only for unfavorable results. A cluster-influence analysis and more independently supported manipulated groups would strengthen inference, but neither was retrospectively manufactured. The present intervals remain the documented conditional analysis, not a certification of population-wide certainty.

5.8. Relationship to Previous Studies

The positive C40 recurrent comparison is compatible with the motivation for richer video representations, but does not refute recurrent approaches as a family. The present ResNet-18/LSTM baseline differs from the architectures and training settings of Güera and Delp (2018) and B. Chen et al. (2022). Likewise, the implemented Xception results reflect a common aligned-cache adaptation and fixed local training procedure. Their weaker FF++ scores cannot establish that all spatial methods fail to capture transferable evidence. The local experiment evaluates selected systems, whereas the RRL describes a much broader design space.

The RQ4 result differs from the positive cross-attention ablations reported for deepfake CAST. In its revised primary preprint, CAST compares CNN/Transformer cross-attention with removed or decoupled attention and reports improved performance; the same study reports that its multiscale variant underperforms its main configuration (Thakre et al., 2026a, Sections 5.5.1–5.5.4). Our null incremental finding concerns fixed Swin/Video-Swin features and a capacity-matched joint readout. The studies therefore do not estimate the same intervention. Their combined implication is conditional: neither cross-attention nor multiscale aggregation has a benefit guaranteed independently of representation, control design, and optimization. The present result does not invalidate CAST's evidence, and CAST's evidence does not establish an unobserved benefit for TLBG-HCA.

Temporal-focused literature also supplies an alternative to adding capacity. FTCN limits spatial convolution while retaining temporal modeling, and LipForensics builds on representations learned from natural mouth motion (Haliassos et al., 2021; Zheng et al., 2021). AltFreezing changes how spatial and temporal weights are optimized to reduce reliance on only one cue family (Wang et al., 2023). Our high within-dataset scores and weaker transfer are compatible with the concern motivating these studies: availability of temporal computation does not ensure use of transferable temporal evidence. They do not, however, prove that SwinFusion++ relied on shortcuts; no cue-removal intervention or causal attribution analysis was performed here. Training objectives and pretraining information remain plausible explanatory factors, not experimentally identified causes.

The comparison policy follows the methodological concern highlighted by DeepfakeBench: preprocessing, partitions, training settings, and scoring units can change the meaning of a reported result (Yan et al., 2023). Accordingly, published CAST, TSFF-Net, and SFormer values were not merged with local measurements into a common leaderboard. In particular, the retained evidence does not authenticate the metric and protocol details needed for a numerical SFormer comparison. Its published architecture remains relevant context, but its scores cannot substantiate local superiority. A valid numerical comparison would require a compatible implementation or a fully reconciled evaluation protocol.

Table 10. Probability quality before and after validation-fitted temperature scaling, for fusion seed 1701 only.

Evaluation	Temperature	Threshold	Brier raw	Brier scaled	ECE raw	ECE scaled
C23 (within)	1.8251	0.6629	0.0229	0.0211	0.0266	0.0295
C40 (within)	3.3636	0.7294	0.0455	0.0430	0.0460	0.0493
Celeb-DF (within)	5.8923	0.2907	0.1218	0.1203	0.1279	0.2447
DeeperForensics (within)	2.2448	0.8367	0.0053	0.0051	0.0057	0.0073
C40 (transfer)	1.8251	0.6629	0.1296	0.1185	0.1496	0.1445
Celeb-DF (transfer)	1.8251	0.6629	0.0979	0.0937	0.0736	0.0730
DeeperForensics (transfer)	1.8251	0.6629	0.6362	0.5815	0.6815	0.6632

Smaller Brier/ECE values are preferable; ECE uses 15 equal-width fake-probability bins. Transfer rows inherit the C23 validation operating point. DeeperForensics transfer is the full overlap-qualified view. These single-seed values are not ensemble calibration estimates.

Table 11. Archived network-only inference profile on two RTX 5090 workers, bfloat16, batch one per device.

Model	Parameters (M)	Clip (ms)	Video (ms)	Dual-GPU clips/s	Peak alloc. (GiB)
Swin-B	86.74	11.51	34.46	172.84	0.599
Video Swin-B	87.64	14.08	42.10	140.56	0.813
SwinFusion++	180.89	28.06	84.11	70.77	1.204
Xception	20.81	7.86	23.72	253.46	0.421
CNN–LSTM	11.97	1.56	4.62	1277.86	0.177

Times are per-device means; a video comprises three sequential anchors. Throughput aggregates both workers. Memory is maximum allocated across workers. Measurements exclude decoding, face processing, input preparation, and multi-seed ensemble execution; principal models use the selected C23 seed-1701 profiles.

Finally, Deepfake-Eval-2024 documents a broader gap between curated benchmarks and collected contemporary media (Chandra et al., 2026). Its acquisition process and multimodal scope differ from the face-centered task here; it is supporting context for external-validity caution, not an additional dataset evaluated by this study. The present contribution is therefore the implementation-grounded assessment of a specific aligned dual-Swin design, including evidence against a stronger mechanism claim, rather than a general solution to all deepfake detection.

5.9. Computational Cost and Practical Implications

The measured resource profile in Table 11 places predictive findings in context. The complete fusion detector had 180.89 million parameters, compared with 86.74 million for Swin-B and 87.64 million for Video Swin-B. Mean clip latency was 28.06 ms versus 11.51 and 14.08 ms, respectively; peak allocated memory was 1.204 GiB versus 0.599 and 0.813 GiB. Relative to Swin-B, fusion required approximately 2.44 times the clip latency. Dual-GPU clip throughput fell from 172.84 to 70.77 clips/s. These are material costs when improved discrimination over the constituent encoders is not consistently established.

The timing boundary matters. Fusion's mean three-anchor video-network time was 84.11 ms per device, but excluded decoding, tracking, alignment, input preparation, and execution of all seed-ensemble members. The reported dual-GPU throughput is not single-device throughput, and repeatedly timing one network-ready input is not a distribution of end-to-end latency over arbitrary videos. Consequently, the results do not establish a universal real-time deployment claim. Local attention constrains one component of computation without eliminating the cost of two encoders.

In the controlled experiment, late concatenation had mean clip latency 25.63 ms, compared with 28.13 ms for full TLBG-HCA and 27.97 ms for no-exchange. The similar total timing reflects the encoders shared by all three designs; a substantial reduction in head parameters need not proportionally reduce complete-system cost. Neither a smaller head nor a favorable network-only profile resolves preprocessing coverage or transferred threshold failures. The practical implication is to report a joint accuracy–error–resource profile before claiming that the additional exchange machinery is justified.

The absence of raw videos and large checkpoints in the review archive limits additional qualitative or deployment analysis. The saved predictions support error counts, but not input-level diagnoses of why particular videos were misclassified. Attribution to a specific lip region, facial texture, or motion artifact requires the corresponding inputs and an appropriate intervention or validated attribution procedure. The quantitative findings therefore describe predictive behavior without assigning unobserved visual causes.

5.10. Integrated Findings and Limitations

Taken together, the five questions receive distinct answers. RQ1 shows high but condition-dependent principal discrimination, without consistent superiority over standalone Swin variants. RQ2 establishes local cross-family evidence for the six verified baseline cells, not the blocked families. RQ3 supplies a substantial, protocol-eligible C40 seed-1701 advantage over the representative CNN–LSTM. RQ4 provides no observed incremental AUC/F1 advantage over the capacity-matched no-exchange head under the single-seed fixed-feature regime. RQ5 reveals weaker frozen transfer, score/threshold and calibration failures, high but incomplete preprocessing coverage, and additional computational cost.

The contribution is therefore not a claim that complexity necessarily improves detection. It is a controlled separation of three propositions: a system can discriminate well within a benchmark; its exchange component may add no detectable value beyond a strong joint control; and its source operating point can still fail on another population. This separation is scientifically useful because it prevents a large baseline improvement from being recycled as evidence for a mechanism that the corresponding intervention did not support. Positive, negative, and inconclusive findings all constrain the thesis.

Several limits govern that interpretation. Training amendments and different nominal budgets restrict architecture attribution; two or three seeds characterize only limited optimization variation. The single-seed frozen-backbone comparison cannot establish end-to-end potential or the separate effects of alignment, locality, stage access, gate initialization, or pretraining. Celeb-DF's custom split and dominant dependency component constrain comparisons with standard test protocols and the generality of its intervals. DeeperForensics is a class-imbalanced, lineage-linked binary construction; the strict subset addresses one documented dependency, not every possible relationship. No formal equivalence margin, prospective power study, or independent relabeling was available.

These limitations identify the scope of the completed evidence rather than licenses to replace it with more favorable assumptions. Additional multi-seed matched-budget training, component interventions, target-lineage audits, and independent-group calibration studies would be needed to strengthen the unresolved claims. They are future tests, not results of this manuscript. Within the retained evidence, SwinFusion++ is best understood as a technically specified and auditable dual-stream detector whose benchmark discrimination is established more strongly than its exchange-specific advantage or its general-purpose forensic reliability.

6. Conclusions

This study evaluated whether face-aligned, hierarchical integration of Swin-B and Video Swin-B provides useful deepfake discrimination and whether that usefulness can be attributed specifically to TLBG-HCA. Its central finding is that within-benchmark discrimination, an incremental advantage from cross-stream exchange, and reliable decisions after distribution shift are separate empirical outcomes. The completed evidence supports the first more strongly than the latter two. The contribution is therefore an implemented and auditable dual-stream configuration together with a bounded assessment of its benefits, rather than a general demonstration that increasing fusion complexity improves forensic reliability.

With respect to the discrimination objective, the homogeneous fusion ensembles obtained video-level ROC-AUCs of 0.9965 on FaceForensics++ DeepFakes C23, 0.9854 on C40, 0.9724 on Celeb-DF v2, and 0.9990 on the retained DeeperForensics binary construction (Table 4). These results establish strong ranking on the analyzed within-dataset populations, but not consistent superiority over the standalone encoders. The Celeb-DF fusion ensemble illustrates the distinction: higher fake-class F1 and accuracy accompanied lower AUC and balanced accuracy than Swin-B, reflecting different false-positive and false-negative trade-offs. No protocol-eligible internal comparison established a positive fusion AUC or F1 advantage after the retained multiplicity correction. Small within-cohort seed variation described only two or three optimization runs; it did not establish stability across new identities, training budgets, or data-generating conditions.

The cross-family objective produced a more specific positive finding. On 280 common C40 test videos, seed-1701 fusion exceeded the representative CNN–LSTM by 0.1846 in AUC and 0.1867 in F1, with Holm-adjusted $p = 0.0003999$ for both endpoints (Table 6). The corresponding paired intervals were positive. This finding supports the specified trained fusion system relative to that implemented recurrent comparator, not an architecture-only causal effect or a conclusion about all recurrent detectors. Differences in pretraining, capacity, and nominal training budgets remain relevant even for an eligible paired test. Similarly, the verified Xception evaluations provide local comparison evidence, whereas the blocked I3D and SFormer reproductions and protocol-qualified literature comparisons cannot substantiate additional numerical superiority claims.

The direct mechanism test did not establish the expected incremental benefit of cross-attention. Under fixed C23 backbone features, full TLBG-HCA and the capacity-matched no-exchange head both achieved AUC 0.9964, F1 0.9857, and accuracy 98.57%; late concatenation achieved AUC 0.9952 and F1 0.9892 (Table 8). Neither full-versus-control contrast established an AUC or F1 improvement under the retained procedure. Because the no-exchange control still combines both streams and their interaction features, the conclusion concerns cross-stream attention versus independent within-stream attention, not fusion versus no fusion. Identical observed metrics and decisions also do not establish equivalent models: the saved logits differed. This single-seed, frozen-feature experiment leaves end-to-end co-adaptation and the separate effects of alignment, locality, hierarchy, and gate initialization unresolved. It nevertheless prevents the positive recurrent-baseline result from being used as evidence for an exchange-specific benefit that the controlled comparison did not demonstrate.

The transfer objective exposed the clearest boundary on generalization. With C23 weights, calibration, and threshold unchanged, fusion AUC declined to 0.9686 on C40, 0.8730 on Celeb-DF, 0.8021 on the full DeeperForensics target, and 0.7682 on its strict lineage-filtered subset (Table 9). On the full DeeperForensics construction, fake recall was only 13.41%, despite an AUC indicating residual ranking information; recall was 17.90% in the strict subset, which likewise did not exhibit dependable decisions. The full and strict views differ in composition and cannot be used to estimate a causal effect of lineage overlap. Both frozen-transfer views exhibited weaker ranking and operating behavior than the target-trained evaluation. A different threshold could change the error balance, but would not itself recover the lost ranking, and target-label-based retuning was not part of the completed experiment.

Calibration, coverage, and dependence further qualify the findings. Temperature scaling slightly reduced within-dataset

Brier error for fusion seed 1701 while increasing its ECE in every retained condition; source-fitted scaling also left substantial probabilistic error on transferred DeeperForensics inputs. Thus, a validation-fitted calibration transformation is not sufficient evidence of reliable target probabilities. Classification performance remained conditional on successful preprocessing: all C23, C40, and Celeb-DF test videos were analyzed, compared with 2,385 of 2,412 full DeeperForensics videos and 354 of 360 strict-subset videos. Moreover, Celeb-DF's largest dependency component contained 814 of 859 test videos, including all manipulated examples. Its narrow conditional intervals therefore cannot establish generalization across many independently represented manipulated groups.

The computational objective also argues against treating predictive scores in isolation. The complete fusion model contained 180.89 million parameters and required 28.06 ms per network-ready clip, compared with 11.51 ms for Swin-B on the recorded profiling configuration. Its 84.11 ms three-anchor video time excluded decoding, face processing, input preparation, and multi-seed ensemble execution. These measurements document an additional resource requirement, not end-to-end real-time readiness. Given the absence of a consistent constituent-encoder advantage and the unestablished exchange benefit, the completed evaluation does not yet demonstrate that this additional cost is offset by a reproducible mechanism-specific improvement.

The scope of these conclusions remains the retained visual, face-centered benchmarks and documented training regimes. Training amendments, unequal budgets, limited seeds, concentrated dependency groups, inherited labels, and the lineage-linked DeeperForensics construction restrict broader attribution. The unresolved claims require matched-budget multi-seed comparisons, targeted component interventions, independently supported target populations, and evaluation of the complete processing pipeline. Those are subsequent tests, not findings supplied by the present study. Overall, SwinFusion++ provides evidence that a well-specified dual-stream detector can achieve strong benchmark discrimination while leaving the value of its exchange mechanism and the reliability of its transferred decisions unproven. The resulting contribution is both architectural and evaluative: it identifies what the implemented integration achieves, what simpler controls already explain, and where the evidence stops.

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